

TESTING, REFINEMENT & VALIDATION RECORD

HealthConnect Experience Lab — Data Analytics Track

HC-POD 8

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1. KPI Cards Validation

Field	Record
Component Tested	5 KPI Cards
Testing Objective	Verify that the KPI values displayed in Power BI match independent calculations from the underlying dataset.
Test/Scenario	Independently recalculate the five dashboard KPI values and compare them with the values displayed in Power BI.
Expected Result	Each KPI value should match the independent calculation.
Actual Result	All five KPI values matched the independent calculations: Overall No-Show Rate 48.5%, No Reminder 51.4%, Previous No-Show 55.4%, 3+ Week Lead Time 56.9%, and Highest Observed Segment 64.1%.
Pass/Fail	PASS
Issue Identified	No issue identified.
Action Taken	No correction required.
Retest Result	KPI values remained correct after dashboard refinement.
Collaborating Track	Data Analytics
Evidence	Python independent calculations, Power BI dashboard screenshot, testing record

2. Booking Lead Time Visual

Field	Record
Component Tested	No-Show Rate by Booking Lead Time
Testing Objective	Verify that the visual displays the correct No-Show Rates for each lead-time group.
Test/Scenario	Compare Power BI values with independent calculations for Same week, 2–3 weeks, and 3+ weeks.
Expected Result	Dashboard values should match the independent calculations: 27.8%, 37.2%, and 56.9%.
Actual Result	All three values matched the independent calculations.
Pass/Fail	PASS
Issue Identified	No calculation issue identified.
Action Taken	No correction required.
Retest Result	Values remained correct after dashboard refinement.
Collaborating Track	Data Analytics
Evidence	Python calculations and Power BI visual

3. Previous No-Show Visual

Field	Record
Component Tested	No-Show Rate by Previous No-Show History
Testing Objective	Verify that the dashboard correctly represents No-Show Rates based on previous No-Show history.
Test/Scenario	Compare Power BI results with independent calculations.
Expected Result	Values should match 43.5% for no previous No-Show and 55.4% for previous No-Show.

Field	Record
Actual Result	Dashboard values matched the independent calculations.
Pass/Fail	PASS
Issue Identified	No issue identified.
Action Taken	No correction required.
Retest Result	Values remained correct.
Collaborating Track	Data Analytics
Evidence	Python calculations and Power BI visual

4. Distance Visual

Field	Record
Component Tested	No-Show Rate by Distance to Clinic
Testing Objective	Verify that distance-group No-Show Rates are correctly displayed.
Test/Scenario	Compare dashboard values with independent calculations for the three distance groups.
Expected Result	Values should match 46.5%, 47.3%, and 54.1%.
Actual Result	Dashboard values matched the independent calculations.
Pass/Fail	PASS
Issue Identified	No issue identified.
Action Taken	No correction required.
Retest Result	Values remained correct after refinement.
Collaborating Track	Data Analytics
Evidence	Python calculations and Power BI visual

5. Reminder Status Visual

Field	Record
Component Tested	No-Show Rate by Reminder Status
Testing Objective	Verify that No-Show Rates for reminder status are correctly displayed.
Test/Scenario	Compare dashboard values for Reminder Sent and No Reminder with independent calculations.
Expected Result	Values should match 47.4% for Reminder Sent and 51.4% for No Reminder.
Actual Result	Dashboard values matched the independent calculations.
Pass/Fail	PASS
Issue Identified	No issue identified.
Action Taken	No correction required.
Retest Result	Values remained correct.
Collaborating Track	Data Analytics
Evidence	Python calculations and Power BI visual

6. Reminder Channel Visual

Field	Record
Component Tested	No-Show Rate by Reminder Channel
Testing Objective	Verify that the No-Show Rates for Email, SMS, and WhatsApp are correctly displayed.
Test/Scenario	Compare Power BI channel results with independent calculations.
Expected Result	Dashboard values should match the independently calculated channel rates.

Field	Record
Actual Result	Dashboard values matched the independent calculations.
Pass/Fail	PASS
Issue Identified	No issue identified.
Action Taken	No correction required.
Retest Result	Values remained correct.
Collaborating Track	Data Analytics
Evidence	Python calculations and Power BI visual

7. Combined Lead Time × Previous No-Show Visual

Field	Record
Component Tested	Combined Lead Time × Previous No-Show visual
Testing Objective	Verify that all six segment No-Show Rates are correctly calculated and displayed.
Test/Scenario	Compare Power BI results with independent calculations across the six combinations of lead time and previous No-Show history.
Expected Result	Dashboard values should match the independent calculations.
Actual Result	All six segment values matched: 21.8%, 36.2%, 34.1%, 41.9%, 51.7%, and 64.1%.
Pass/Fail	PASS
Issue Identified	No calculation issue identified.
Action Taken	No correction required.
Retest Result	Calculations remained correct after refinement and filtering tests.

Field	Record
Collaborating Track	Data Analytics
Evidence	Python calculations and Power BI visual

8. Appointment Type Slicer

Field	Record
Component Tested	Appointment Type Slicer
Testing Objective	Determine whether the slicer correctly filters the relevant dashboard visuals.
Test/Scenario	Select different appointment types and observe the dashboard response.
Expected Result	Relevant KPI cards and visuals should update according to the selected appointment type.
Actual Result	Dashboard responded correctly to the selected appointment types.
Pass/Fail	PASS
Issue Identified	No issue identified.
Action Taken	No correction required.
Retest Result	Filtering continued to work correctly after refinement.
Collaborating Track	Data Analytics
Evidence	Power BI slicer testing and screenshots

9. Gender Slicer

Field	Record
Component Tested	Gender Slicer

Field	Record
Testing Objective	Determine whether the Gender slicer correctly filters the dashboard results.
Test/Scenario	Test Female, Male, and Prefer not to say selections and compare selected results with independent calculations.
Expected Result	Dashboard results should update correctly and match independent calculations.
Actual Result	Dashboard filtering worked correctly, and the tested results matched independent Python calculations.
Pass/Fail	PASS
Issue Identified	The Prefer not to say group showed a reversed pattern in the 2–3 weeks segment.
Action Taken	The result was documented as a small-sample caution rather than treated as a general behavioural pattern. No dashboard calculation was changed.
Retest Result	Filtering and calculations remained correct.
Collaborating Track	Data Analytics
Evidence	Power BI testing, Python calculations, and testing notes

10. Lead Time Band Ordering

Field	Record
Component Tested	Lead Time Band ordering
Testing Objective	Verify that lead-time categories appear in the intended chronological order.
Test/Scenario	Test the ordering of Same week, 2–3 weeks, and 3+ weeks in the Power BI visuals.
Expected Result	Categories should follow the chronological order: Same week → 2–3 weeks → 3+ weeks.

Field	Record
Actual Result	The previous sort configuration produced an ordering issue caused by a circular dependency in the sort-by-column setup.
Pass/Fail	FAIL → PASS
Issue Identified	Lead Time Band sort-order issue.
Action Taken	Created a separate Lead Time Band Sort column directly from booking_lead_days and applied it using Sort by Column.
Retest Result	The categories remained in the intended chronological order after refinement.
Collaborating Track	Data Analytics
Evidence	DAX Lead Time Band Sort calculation, Power BI screenshots, before/after testing

11. Data Science Segment Calculation

Field	Record
Component Tested	Data Science Lead Time × Previous No-Show segment calculation
Testing Objective	Independently verify the segment calculation reported by the Data Science track.
Test/Scenario	Reproduce the exact Data Science definition: booking_lead_days > 30 and previous_no_shows ≥ 1, using both denominator definitions.
Expected Result	Independent calculation should reproduce the reported Data Science results.
Actual Result	The calculation reproduced 70.52% when Cancelled appointments were excluded and 67.88% when Cancelled appointments were included.
Pass/Fail	PASS
Issue Identified	The original analyses used different lead-time thresholds and denominator definitions.

Field	Record
Action Taken	The segment criteria and denominator definitions were explicitly documented for future cross-track comparisons.
Retest Result	The Data Science calculation was independently reproduced exactly under its stated definitions.
Collaborating Track	Data Science — Donald Nwachukwu
Evidence	Python independent calculation and Data Science output/message

12. Dashboard After Refinement

Field	Record
Component Tested	Refined Power BI Dashboard
Testing Objective	Confirm that the Lead Time Band refinement did not introduce new calculation or filtering errors.
Test/Scenario	Re-test the 5 KPI cards, 6 comparison visuals, 2 slicers, combined visual, and Lead Time Band ordering.
Expected Result	Previously validated values and filtering behaviour should remain unchanged after refinement.
Actual Result	All tested components continued to function correctly. No new calculation or filtering issues were identified.
Pass/Fail	PASS
Issue Identified	No new issue identified.
Action Taken	No further correction required.
Retest Result	Overall re-test: PASS.
Collaborating Track	Data Analytics

Field	Record
Evidence	Before/after screenshots, Python calculations, Power BI dashboard, testing record

Overall Testing Result

The Week 7 testing confirmed that the main **KPI cards, comparison visuals, dashboard slicers, combined analysis, and analytical calculations** were consistent with independent calculations.

One dashboard presentation issue involving **Lead Time Band ordering** was identified, corrected using a separate Lead Time Band Sort column, and successfully retested.

Cross-track testing with **Data Science** also independently reproduced the reported segment calculation and clarified that the different reported No-Show Rates resulted from differences in the **lead-time threshold and denominator definition**.

Overall Week 7 Testing Status: PASS